

Received 23 August 2024, accepted 11 September 2024, date of publication 17 September 2024,
date of current version 30 September 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3462547

RESEARCH ARTICLE

An Image Double Encryption Based on Improved GAN and Hyper Chaotic System

LONGXUAN DAI, LEI HU^{ID}, (Member, IEEE), LIANGBING CHEN,
CHENXI WANG, AND FENG LIN

School of Computer Information Engineering, Jiangxi Normal University, Nanchang 330022, China

Corresponding author: Lei Hu (hulei@jxnu.edu.cn)

This work was supported by the National Natural Science Foundation of China under Grant 62362040.

ABSTRACT In the current research field of digital image encryption technology, there are some key challenges, including insufficient key space, the weak sensitivity of key parameters, and low complexity of the encryption process. To address these issues, this study proposes a double image encryption framework to enhance the security of image encryption algorithms. In the first encryption, this study proposes an improved generative adversarial network incorporating unsupervised learning properties, adaptive layer instance normalization, multi-level attention enhanced residual network structure, and feature distribution fidelity module. This study also proposes a comprehensive loss function for the network, which consists of an encryption loss function, a decryption loss function, an image space smoothness improvement loss function, a generator veracity enhancement loss function, and an identity loss function. In the second encryption, this study proposes a six-dimensional hyperchaotic system possessing four positive Lyapunov exponents for the first scrambling operation, and combines the Fibonacci Q-matrix and classical shuffling algorithms for the diffusion and second scrambling operations. By combining deep learning techniques with chaotic systems, the key space and the quality of the encrypted images are effectively improved, as well as the ability of resistance to the burst attack, salt-and-pepper noise attack and cropping attack. This study provides new perspectives and technical routes for solving the challenges faced in digital image encryption.

INDEX TERMS Image encryption, generative adversarial networks, hyperchaotic systems, deep learning, encryption and decryption.

I. INTRODUCTION

With the rapid development of the digital era, the role of image data in our daily lives and many industries is becoming more and more significant, involving a variety of fields such as healthcare, military security, media communication, and personal privacy protection. Although the wide application of image data brings convenience, it also increases the risk of data theft and forgery [1]. People pay more and more attention to the confidentiality, authenticity, security, privacy, and integrity of sensitive image data [2]. Image encryption techniques aim to secure the transmission and storage of image data in non-secure network environments. Image encryption effectively prevents unauthorized access by transforming image data into a format that can only be

interpreted by authorized users. Compared to traditional text encryption techniques, image encryption faces more complex challenges in handling large amounts of data, high data relevance, and redundancy. And image information can convey information more intuitively and vividly. But this also exposes various security risks [3]. Therefore, the development of efficient and secure image encryption algorithms has become an urgent and critical task in this field.

Chaotic systems are widely used in image encryption research due to their ability to generate secure key sequences and ease of implementation [4]. Chaotic systems include one-dimensional and multi-dimensional chaotic systems. Chaotic systems can be controlled with few parameters and hence are widely used in different image encryption algorithms [5]. Khairullah et al. [6] proposed an efficient and fast encryption method. The method combines one-dimensional improved logistic mapping (1D-ILM) and

The associate editor coordinating the review of this manuscript and approving it for publication was Jiachen Yang^{ID}.

one-dimensional improved quadratic mapping (1D- IQM). Zhu et al. [7] proposed a one-dimensional (1D) piece-wise quadratic polynomial chaotic map (PWQPCM) image encryption method with the advantages of low time cost and flexible security strength adjustment. However, the one-dimensional chaotic system has the disadvantages of simple structure, uncomplicated chaotic behavior, and narrow chaotic range [8], which leads to the generation of key sequences with insufficient complexity and easy to be cracked. In order to break through the limitation of 1D chaotic mapping in image encryption, many researchers have combined 1D chaotic systems with methods in other fields. Liang et al. [9] proposed an image encryption algorithm based on one-dimensional sinusoidal cosine chaotic mapping (SCCM) and random DNA coding to address the shortcomings of one-dimensional chaotic systems. The implementation of row and column index misalignment of plaintext images enables the algorithm to achieve higher security and complexity. Rehman [10] proposed to utilize two-dimensional quantum coding and one-dimensional sinusoidal Chaotic Mapping (1-D SBCM) to address the limitations of pseudo-random and periodic properties in traditional 1-D chaotic systems. In addition, some researchers proposed new multidimensional chaotic systems to increase the complexity of the image encryption process by generating complex pseudo-random sequences. Ye et al. [11] proposed an asymmetric image encryption scheme based on Improved Lorenz System (ImproLS) and undefined signature. The method cleverly exploits the strong chaotic property of the Improved Lorenz System and improves the security and attack resistance of image encryption by combining the Arnold transform, RSA algorithm and discrete wavelet transform. Considering the necessity of authentication in most scenarios, Guo et al. [12] proposed a dual-channel encryption algorithm combining digital and optical channels. Using computational ghost imaging (CGI) technique, the encryption process is controlled by pseudo-random sequences generated by a 4D chaotic system and Logistic maps. For multi-image encryption, Zhou et al. [13] proposed a multi-image encryption scheme using quaternion discrete fractional Tchebyshev moment transform (QDFrTMT). The scheme employs the Logistic-sine exponential chaotic map and piece-wise linear chaotic map to enhance security. Images are scrambled and then encrypted using QDFrTMT with dual-layer protection. Es-Sabry et al. [14] used the DNA encoding technique with Baker map based on an improved Rossler chaotic system. This innovative approach solves the problem that traditional encryption algorithms such as RSA are unable to encrypt multimedia data effectively. Wang et al. [15] proposed a chaos-based image encryption algorithm for non-regular-sized specific content. The method innovatively constructs a novel LS chaos model combining sinusoidal and logarithmic maps, and improves its chaotic complexity by perturbing the control parameters and introducing delayed states. Huang and Cai [16] proposed a dual-color image encryption

method for IIoT for the massive data in the era of industrial IoT. The study utilizes the three-dimensional non-equilateral Arnold transform (3D-NEAT) for the disambiguation operation, and the random diffusion sequence generated by the three-dimensional Lorenz system (3D-LS) for the diffusion and the second disambiguation operation to obtain the final encrypted image. Multidimensional chaotic systems are also capable of generating more complex chaotic mappings when they have multiple positive Lyapunov exponents [17]. Li et al. [18] proposed an image encryption method based on a newly designed 5D fractional-order hyperchaotic system to address the problems of low efficiency and vulnerability to cracking in the existing image encryption techniques. Unlike the traditional point-by-point diffusion method, this method combines fast row- and column-level joint substitution and diffusion operations by changing the pixel values row-by-row or column-by-column. Neamah et al. [19] proposed a grey-scale image encryption method combining a seven-dimensional hyperchaotic system with Pascal's matrix for scrambling operations on the original image. The method overcomes the limitations imposed by low-dimensional chaotic systems by using hyperchaotic systems and provides a method to encrypt grey-scale images efficiently. Ning et al. [20] proposed a method to encrypt any greyscale image and color image by combining a novel hyperchaotic system and a newly designed bit plane segmentation method. The method constructs a five-dimensional Hamiltonian hyperchaotic system through Euler's equation and energy analysis theoretical basis and combines it with the bit plane diffusion method at the bit level. In addition, the combination of biology [21], [22], [23], signal processing [24], [25], [26] and optics field [27], [28] with chaotic systems explores a new direction in the research of image encryption algorithms. The integration of cross-domain technologies further enhances the complexity of encryption algorithms and increases the difficulty of cracking by attackers [29].

With further research in deep learning, deep learning is gradually being used in various fields. Multi-layer convolutional networks can extract multi-layer image feature maps in the input image. Convolutional networks have been widely used in the fields of target detection, image super-resolution and image style migration, which have been proven to have significant advantages. Through the learning ability of deep neural networks, images can be effectively encoded, greatly increasing the difficulty of unauthorized decoding and ensuring the security of sensitive information. At the same time, the fast parallel processing and matrix operation characteristics of neural networks make them well-suited for image encryption [30]. Some researchers have begun to explore the direct use of deep neural networks to achieve end-to-end image encryption. Chai et al. [31] designed thumbnail preserving encryption (TPE) based on sum-preserving encryption. CycleGan is used to simulate Random Unitary Encoding (RUE) and achieve more efficient TPE. Meanwhile, the method of thumbnail consistency loss is

proposed and the quality of decrypted images is improved by the loss function. Wang and Lo [32] proposed an image compression framework incorporating deep learning. An autoencoder-based network is used for image compression and the extracted deep features are then added to the encryption layer to achieve end-to-end image encryption.

In recent years, in order to further improve the complexity of image encryption methods, there are many researches combine that combine deep learning with traditional encryption methods to propose image encryption techniques [33], [34] and [35], which take full advantage of deep learning and traditional encryption methods. Liu and Xue [36] proposed an image encryption method that combines generative adversarial networks and compression perception. First, the image is sparsified by discrete wavelet transform and compression techniques. Then, the image is rearranged and diffused using an improved Henon mapping technique to generate a ciphertext image. Finally, the encrypted image is converted back to the visually meaningful original image by the generative adversarial network. Long et al. [30] proposed a deep learning-based encryption method for medical images. The key generated using logistic mapping is combined with image features to improve the diffusion performance of the encryption process, a feedback neural network is used to improve the reconstruction quality of the image, and a new loss function is designed to evaluate the encryption effect and optimize the network. Abdellatif et al. [37] proposed an improved CT image encryption method for privacy preservation of COVID-19 diagnosis. The method uses a convolutional neural network to extract features from face images and applies these features to chaotic mapping to generate encrypted CT images.

This study proposes a double encryption-decryption framework for images. The first encryption utilizes an improved generative adversarial network for end-to-end encryption operations. The plaintext image is merged with an image whose pixels have Gaussian distribution as input. The second encryption uses the biometric fingerprint to generate the initial value of the pseudo-random sequence for the six-dimensional hyperchaotic system, and the chaotic system iterates to generate the pseudo-random sequence. Moreover, the Fibonacci Q-matrix and Knuth-Durstenfeld Shuffle algorithm are combined to further scrambling and diffuse the intermediate ciphertext image to enhance the encryption effect. The decryption process is the inverse process of the encryption operation. The image is decrypted by the corresponding inverse operation and the trained decryption network.

Overall, the main contributions of the work are summarised below:

(1) This study proposes an improved generative adversarial network. It incorporates the unsupervised learning properties of Cycle Generative Adversarial Network (CycleGAN) and Adaptive Layer Instance Normalisation (AdaLIN). Meanwhile, the Multi-level Attention Enhanced

Residual Network Structure (MAERS) and Feature Distribution Fidelity Module (FDFM) are constructed.

(2) In the improved generative adversarial network, this study designs a comprehensive network loss function. This loss function incorporates an encryption loss function, a decryption loss function, an image space smoothness improvement loss function, a generator veracity enhancement loss function, and an identity loss function.

(3) This study constructs a six-dimensional hyperchaotic system with four positive Lyapunov exponents in the second encryption. The initial value of the chaotic system is determined by the biometric fingerprint feature, which ensures that the encryption process has a unique biometric identifier.

(4) Based on the comparative experiments and the analysis of relevant indicators, the image encryption and decryption method proposed in this study performs well in a variety of security indicators and can effectively defend against multiple forms of attacks.

In the rest of the paper, in Section II, the first encryption process based on improved generative adversarial networks is presented. In Section III, the second encryption process based on a six-dimensional chaotic system, Fibonacci Q-matrix and shuffling algorithm is presented. In Section IV, the evaluation of encryption and decryption effects and simulation experiments are carried out. In Section V, the proposed research method is summarised.

II. THE FIRST ENCRYPTION PROCESS BASED ON IMPROVED GENERATIVE ADVERSARIAL NETWORK

A. THE FIRST IMAGE ENCRYPTION AND DECRYPTION PROCESS

In this section, the training process of the network in our proposed first heavy encryption is described in detail. Then, the improved network structure and loss function are presented through diagrams, tables and formulas. In the first encryption, this study employs an improved generative adversarial network, which ensures that the key space is large enough. The experimental results show that the encryption and decryption networks are mutually supportive. Each training session produces different network parameters and thus different encryption and decryption keys. This means that even if the attacker has the network structure, he still cannot decrypt it to obtain the plaintext information in the absence of training parameters.

This innovation allows to further enhance the key sensitivity of the encryption process. The first encryption and decryption process is shown in Figure 1. The plaintext image and a Gaussian random noise image undergo channel merging to serve as an input to the image double encryption framework. Assume that M and N represent the length and width of the image, respectively. When the plaintext grayscale image size is $M \times N \times 1$, the set mean and standard deviation are used to generate a Gaussian random number matrix, and the random number matrix is normalized to the range of 0 to 255 by linear transformation. Through the above

the multi-level feature attention mechanism. This convolutional block configuration is designed to enhance the ability of the network to capture and express key features. The MFAM processing starts with passing the downsampled feature map F_{down} through the global attention module, which compresses the global information into a global information feature vector using adaptive mean pooling to reduce the spatial dimensionality. The global information feature vectors are processed through a sequence of fully connected layers containing two linear transformations and a ReLU activation function using a Sigmoid function to generate global information attention weights. The global information attention weights are used to weight F_{down} to generate the global attention weighted feature maps. At the same time, the downsampled feature map F_{down} is also processed by the local attention module, which identifies the importance of each spatial location by using a convolutional layer and a Sigmoid function, and the resulting local attention weights are also used to weight F_{down} , generating a local attention-weighted feature map. By combining these two weighted feature maps, it is possible to capture both global information and local details, resulting in an information-rich attention fusion feature $F_{attention}$. In addition, F_{down} are also processed by a composite convolution block: two convolutions, two batch normalizations, ReLU activation, and a dropout layer, to obtain the multi-stage processing features. Finally, the multi-stage processing features and the attention fusion features $F_{attention}$ are integrated to generate the output F_{out1} of the convolutional block configuration 1. This output feature effectively integrates global and local information, which improves the performance of the model in complex scenarios.

Convolutional Block Configuration 2 emphasises nonlinear activation processing and flexible padding strategies to optimise feature extraction. Firstly, the output of convolutional block configuration 1 F_{out1} is adjusted to the feature map edges by a fill layer, followed by enhanced nonlinear processing using the ReLU activation function. Next, feature extraction is performed via a 3×3 convolutional layer to ensure that the convolutional padding matches the parameter settings. Subsequently, training is stabilized and convergence is accelerated via a normalization layer. In addition, the residual connection fuses the output of the processing process described above with F_{out1} to obtain the output of the convolution block configuration 2 F_{out2} .

Convolutional block configuration 3 is designed to further enhance feature representation by adjusting the filling strategy and convolutional layer configuration to improve the performance of processing features. Firstly, the spatial dimensions of the input feature map F_{out2} are maintained using a specific reflection method. Subsequently, a 3×3 convolutional layer is used to extract features. Next, a normalization method is used to standardize the features and accelerate the learning and convergence process of the model. Finally, nonlinearities are introduced through the ReLU activation function to enhance the data representation of the model.

The output of the above processing is fused with F_{out2} using residual connection, which is the output of MAERS.

2) FEATURE DISTRIBUTION FIDELITY MODULE

The Feature Distribution Fidelity Module (FDFM) is mainly composed of an Adaptive Normalised Parameter Generator (ANPG), two convolutional layers, two filler layers, two AdaILN layers and one layer of ReLU activation function. In the encryption phase, the FDFM adjusts the feature map to precisely match the feature distribution of the ciphertext. And in the decryption stage, the features are adjusted to reconstruct the feature distribution as close as possible to the plaintext to ensure the authenticity of the generated image. The specific structure of FDFM is shown in Figure 3. Firstly, the input feature maps are border-adjusted by a fill layer to prevent loss of edge information and to maintain consistent feature map size. Subsequently, multiple convolutional layers are processed to extract key information. The first convolutional layer is followed by an adaptive layer instance normalization (AdaILN) and ReLU activation function. AdaILN performs a linear transformation of the normalized feature map using the parameters γ and β generated by ANPG, aiming to ensure that the image generated in the encryption phase conforms to the feature distribution of the ciphertext. At the same time, the image in the decryption phase can also be as close as possible to the original feature distribution of the plaintext. This design not only ensures the confidentiality of the image information in the encryption process, but also improves the restoration of the decrypted image, which effectively improves the performance of the image security task.

C. LOSS FUNCTION

For GAN-based encryption network E and the corresponding decryption network D, we propose an comprehensive network loss function that incorporates encryption loss function, decryption loss function, image space smoothness improvement loss function, generator veracity enhancement loss function, and identity loss function, aiming to comprehensively consider encryption and decryption efficiency, image quality, and accurate identification of key features. The comprehensive network loss function is shown in Eq. (4), as shown at the bottom of page 7, where it is assumed that the plaintext image x is of size $M \times N$, M is the length of the image, and N is the width of the image. $x \sim p_{data}(x)$ is the image feature distribution of the plaintext image. y is the distribution of the Gaussian random noise image with an image size of $M \times N$. $y \sim p_{data}(y)$ is the image feature distribution of the Gaussian random noise image. R denotes the image matrix after the concatenation of the plaintext image and the Gaussian random noise image channel. $R \sim p_{data}(R)$ is the distribution of image matrix features after concatenation of plaintext image and Gaussian random noise image channels. E denotes the encryption network and D denotes the decryption network. m denotes the number of images matching the distribution of

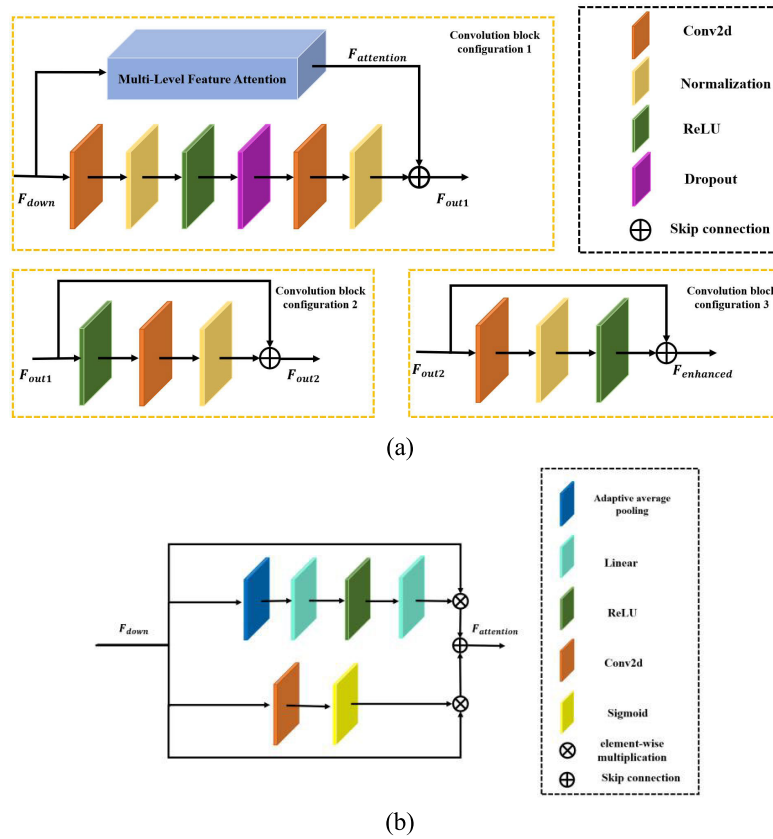


FIGURE 2. Multi-level Attention Enhanced Residual Structure:(a) Convolutional Block Configuration 1, Configuration 2 and Configuration 3 Structure; (b) Multilevel Feature Attention Mechanism.

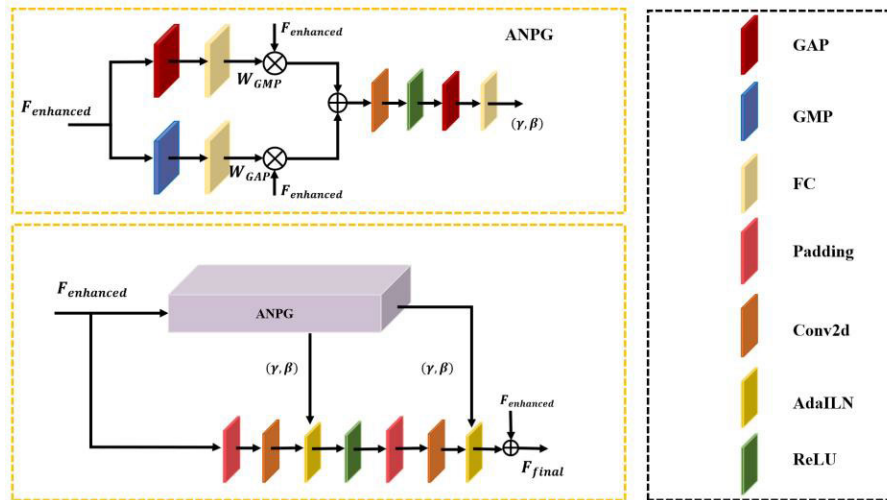


FIGURE 3. Feature distribution FIDELITY module.

image features of the plaintext image. i denotes the index of the image. x_i and y_i denote the i -th corresponding plaintext image and Gaussian random noise image in the training process, respectively. $\mathbb{E}$ denotes the expectation. α , β , μ , φ and γ denote the correlation adjustment parameters ($\alpha = 3.0$, $\beta = 10^{-5}$, $\mu = 3000.0$, $\varphi = 0.3$, $\gamma = 0.1$), and σ denotes the Sigmoid activation function.

1) ENCRYPTION LOSS FUNCTION

The encryption loss function focuses on the generated image to be as similar as possible to the ciphertext image, and the specific encryption loss function is shown in Eq. (5). During the encryption process, the generated image should emulate the feature distribution of the ciphertext image as much as possible, aiming to deceive the eavesdropper or attacker. In

this way, the encrypted network generates outputs that can be indistinguishable from the real ciphertext image, increasing the disorientation of the attacker.

2) DECRYPTION LOSS FUNCTION

The image spatial smoothness improvement loss function helps to improve the overall smoothness and visual effect of an image. The visual experience is improved by inducing a smooth gradation of image brightness. The specific loss function is shown in Eq. (7), as shown at the bottom of the page, where $D(E(R))_{w,h}$ denotes the pixel at position (w,h) of image $D(E(R))$.

3) IMAGE SPACE SMOOTHNESS IMPROVEMENT LOSS FUNCTION

The image spatial smoothness improvement loss function helps to improve the overall smoothness and visual effect of an image. The visual experience is improved by inducing a smooth gradation of image brightness. The specific loss function is shown in Eq. (7), where $D(E(R))_{w,h}$ denotes the pixel at position (w,h) of image $D(E(R))$.

4) GENERATOR VERACITY ENHANCEMENT LOSS FUNCTION

Generator veracity enhancement loss function is a type of loss function designed to enhance the realism of generated images. This loss function is designed to enable the generator to produce images that are more difficult to be recognized by the discriminator by quantifying the realism of the generated images. This loss function captures the combined features of the feature image using GAP and GMP in ANPG. The outputs of GAP and GMP are computed as logical scores through two fully connected layers respectively. Subsequently, these logistic scores are merged along the feature dimensions and the merged logistic scores are compared with the truth labels through Binary Cross Entropy Loss (BCE) to quantify the truthfulness of the generated images. We assume that the

truthfulness labels are all-1 vectors. The specific loss function is shown in Eq. (8), as shown at the bottom of the page, where logit_rec_i represents the logical score of the decrypted image generated by the decryption network D, logit_fake_i represents the logical score of the encrypted image generated by the encrypted image E, and m is the number of image samples.

5) IDENTITY LOSS FUNCTION

Our application to identity loss aims to enhance the model's ability to maintain the features of the input image. When the input image is already in the target domain, the generator can make the output image as similar as possible to the input image. This loss function is effective in the task of preserving image details and avoiding unnecessary distortion during the conversion process. In this study, this loss function ensures that images conforming to the pixel distribution of the ciphertext image are processed by the encrypted network so that the encrypted image can be similar to the pixel distribution of the input ciphertext image. The image that matches the pixel distribution of the plaintext image is processed by the decryption network, and the output image of the decryption network can be similar to the pixel distribution of the input plaintext image. The identity loss function is shown in Eq. (8).

III. SECOND ENCRYPTION PROCESS BASED ON SIX-DIMENSIONAL CHAOTIC SYSTEM, FIBONACCI Q-MATRIX AND SHUFFLING ALGORITHM

A. THE SECOND IMAGE ENCRYPTION AND DECRYPTION PROCESS

This section describes our proposed second encryption process in detail. This section give the details of the methodological techniques and steps of implementation through diagrams and formulas.

In the second encryption process, there are two scrambling operations and one diffusion operation. The workflow of the second encryption is shown in Figure 4. In the first

$$L_{\text{overall}}(E, D) = \alpha L_e(E, D) + L_d(E, D) + \beta L_h(E, D) + \mu L_{GVE}(E, D) + L_{id}(E, D) \quad (4)$$

$$L_e(E, D) = \mathbb{E}_{x \sim p_{\text{data}}(x), y \sim p_{\text{data}}(y)}[(E(x + y) - y)^2] = \mathbb{E}_{x \sim p_{\text{data}}(x)}\left[\sum_{i=1}^m (E(x_i + y_i) - y_i)^2\right] \quad (5)$$

$$L_d(E, D) = \mathbb{E}_{x \sim p_{\text{data}}(x), y \sim p_{\text{data}}(y)}[(D(E(x + y)) - x)^2] = \mathbb{E}_{x \sim p_{\text{data}}(x), y \sim p_{\text{data}}(y)}\left[\sum_{i=1}^m (D(E(x_i + y_i)) - x_i)^2\right] \quad (6)$$

$$L_h(E, D) = \mathbb{E}_{R \sim p_{\text{data}}(R)}\left[\sum_{w=1}^W \sum_{h=1}^H \sqrt{(D(E(R))_{w+1,h} - D(E(R))_{w,h})^2 + (D(E(R))_{w,h+1} - D(E(R))_{w,h})^2 + (D(E(R))_{w+1,h+1} - D(E(R))_{w,h})^2}\right] \quad (7)$$

$$L_{GVE}(E, D) = \left(-\frac{1}{m} \sum_{i=1}^m \log(\sigma(\text{logit_rec}_i))\right) + \left(-\frac{1}{m} \sum_{i=1}^m \log(\sigma(\text{logit_fake}_i))\right) \quad (8)$$

$$L_{id}(E, D) = \varphi \text{loss_id_A} + \gamma \text{loss_id_B} = \mathbb{E}_{x \sim p_{\text{data}}(x), y \sim p_{\text{data}}(y)}[\varphi |E(y + y) - y| + \gamma |D(x) - x|] \quad (9)$$

scrambling operation of the second encryption, this study proposes a six-dimensional hyperchaotic system with high randomness and complexity. The pseudo-random sequence generated by this system is difficult to predict. And based on the uniqueness of biometric fingerprints, it ensures that the initial conditions of each encryption operation are unique and difficult to reproduce; In the second scrambling operation of the second encryption, this study generates random seeds from plaintext images. The plaintext dependency of the encryption process is enhanced to ensure that the encryption keys are different for different plaintext images. It aims to jointly ensure the security and reliability of the encryption process.

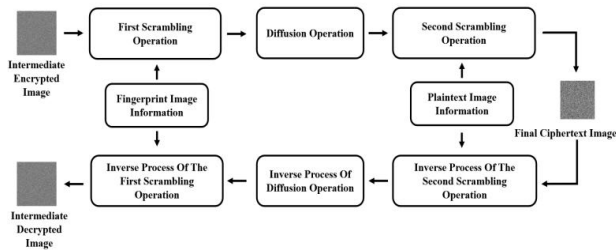


FIGURE 4. The second encryption process based on the six-dimensional chaotic system, Fibonacci Q-matrix, and shuffle algorithm.

In the first scrambling operation, biometric fingerprint images generate the initial values for a six-dimensional hyperchaotic system, subsequently producing six pseudo-random sequences. From these sequences, three sequences are selected and concatenated to create the pseudo-random sequence used for the first scrambling operation. The intermediate ciphertext image from the first encryption serves as the input, employing the concatenated pseudo-random sequence to scramble the image pixels randomly.

In the diffusion operation, the image first scrambling is divided into multiple 2×2 matrices. Subsequently, matrices that exhibit Fibonacci Q-matrix characteristics are used for diffusion. Eight 2×2 matrices, consistent with these characteristics, are selected to perform matrix multiplication with the scrambled image, altering the pixel values sequentially by rows and columns.

During the implementation of the second scrambling operation, the sum of the pixel values from the diffused image is calculated to generate a normalized value. This normalized value serves as the random seed for the Knuth-Durstenfeld Shuffle algorithm, which is used to perform the second scrambling of the image. The specific process starts with the last pixel of the image, randomly selects a position's pixel to swap with the current pixel, moves to the previous pixel, and repeats this process until all pixels are processed, resulting in the final encrypted image.

B. NEW SIX-DIMENSIONAL HYPERCHAOTIC SYSTEM

The design of six-dimensional hyperchaotic systems is based on the extension and improvement of traditional chaotic

systems by introducing more state variables to increase the complexity and stochasticity of the system. A hyperchaotic system is defined as a chaotic system with at least two positive exponentials, which means that its dynamics extends in several different directions at the same time [38]. According to this feature, hyperchaotic systems have more complex dynamical behavior. The presence of multiple positive Lyapunov exponents improves the security of the communication scheme by generating more complex dynamics. At the same time, it can also be used to improve the security of chaotic communication systems behavior [39].

In this study, a six-dimensional hyperchaotic system is used to provide a new method to improve the security of the encryption process. Hyperchaotic systems are highly stochastic and unpredictable, which increases the complexity of encryption algorithms [40]. The existence of multiple Lyapunov exponents will lead to more complex chaotic system dynamics, thus improving the security of information transmission. In addition, the high sensitivity to initial conditions is a characteristic of this kind of chaotic systems, which provides an opportunity for the application of chaotic systems in secure information transmission [41].

In image second encryption and decryption, the first disruption and its inverse operation utilize a chaotic system. This study creates a new six dimensional hyper chaotic system with the formula shown in Eq. (10), where $x_1(0), x_2(0), x_3(0), x_4(0), x_5(0)$ and $x_6(0)$ denote the state variables.

$$\begin{cases} \dot{x}_1 = -\frac{1601}{100}x_1 + 7x_2 - 11x_3 + 10x_4 + 6x_5 + 11x_6 \\ \dot{x}_2 = 2x_1 + \frac{399}{100}x_2 - 16x_3 + 5x_4 + x_5 + 6x_6 \\ \dot{x}_3 = -\frac{79}{5}x_1 + \frac{46}{5}x_2 - \frac{1801}{100}x_3 + \frac{36}{5}x_4 + \frac{31}{5}x_5 + \frac{56}{5}x_6 \\ \dot{x}_4 = -20x_1 + 5x_2 + 3x_3 + \frac{599}{100}x_4 + 2x_5 + 7x_6 \\ \dot{x}_5 = \frac{78}{5}x_1 + \frac{47}{5}x_2 - \frac{53}{58}x_3 + \frac{52}{47}x_4 + \frac{639}{32}x_5 + \frac{37}{1039}x_6 \\ \dot{x}_6 = 17 \sin 9x_1 + x_2 + \frac{5}{5}x_3 + \frac{5}{5}x_4 + \frac{5}{5}x_5 + \frac{1039}{100}x_6 \end{cases} \quad (10)$$

In the encryption method proposed in this paper, the initial state value x_1 of the chaotic system is generated using the biometric fingerprint image B information, and the initial state value of the remaining five dimensions is generated using the initial state quantity x_1 , as shown in Eq. (11) and Eq. (12). And the start point T_1 and end point T_2 of the integration of the chaotic system are determined by Eq. (13) and Eq. (14). Finally, the chaotic sequences $X1, X2, X3, X4, X5$ and $X6$ are obtained through the initial state associated with the biometric fingerprint image. three pseudo-random sequences are extracted from them for collocation, and in this paper, $X1, X2$ and $X5$ are extracted, and from the pseudo-random sequences obtained by collocation of the three pseudo-random sequences, the sequences with the length of $M \times N$ are extracted as the pseudo-random sequences required for the

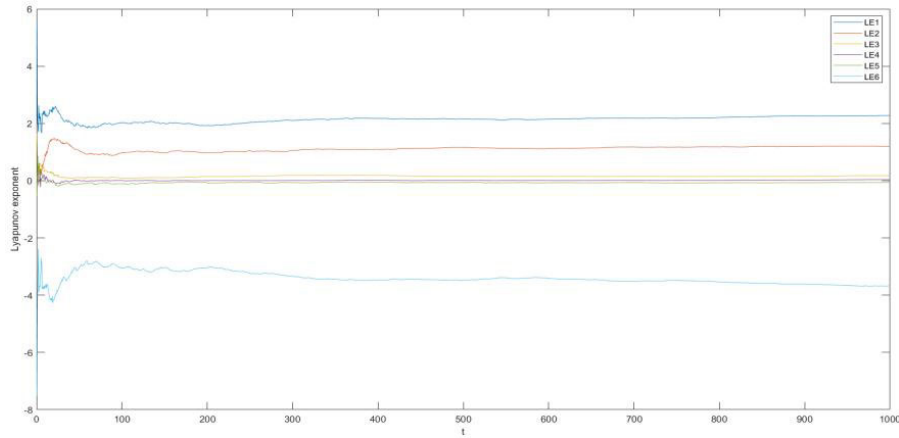


FIGURE 5. Diagram of the Lyapunov Index.

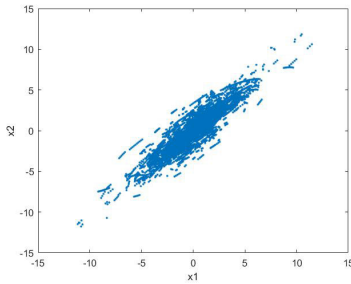
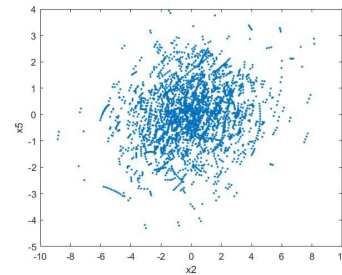
(a) Poincare cross section on (x_1, x_2) plane(b) Poincare cross section on (x_2, x_5) plane

FIGURE 6. Poincare cross sections of chaotic systems.

scrambling operation. Where W and H are the image length and width of the biometric fingerprint image B , respectively. x_i denotes the initial state quantity of dimension i , and the values of i range from 2, 3, 4, 5, 6.

$$x_1(0) = \frac{\text{sum}(B) + W * H}{W * H + 2^{23}} \quad (11)$$

$$x_{i+1}(0) = \text{mod}(x_i \times 10^6, 1) \quad (12)$$

$$T_1 = \frac{0.9865 \times W \times H}{3} \quad (13)$$

$$T_2 = \text{ceil}\left(\frac{5 \times W \times H}{14}\right) \quad (14)$$

1) DISSIPATION OF CHAOTIC SYSTEMS

The dissipation of the chaotic system determines whether there is a chaotic attractor in the system, when the rate of change of the volume of the phase space $\nabla \cdot V < 0$, indicates that the system's motion trajectory will converge to the attractor with a volume of 0, and therefore verifies that the system is chaotic. Evaluating the dissipation of the system according to Eq. (15), we get $\nabla \cdot V < 0$, which proves the existence of attractors of the system. Therefore, the chaotic system is dissipative and is converging to 0 at an exponential

$$\text{rate } \frac{dV}{dt} = e^{-1.05}.$$

$$\nabla \cdot V = \frac{\partial \dot{x}_1}{\partial x_1} + \frac{\partial \dot{x}_2}{\partial x_2} + \frac{\partial \dot{x}_3}{\partial x_3} + \frac{\partial \dot{x}_4}{\partial x_4} + \frac{\partial \dot{x}_5}{\partial x_5} + \frac{\partial \dot{x}_6}{\partial x_6} \quad (15)$$

2) LYAPUNOV EXPONENTS AND DIMENSIONAL ANALYSIS

The Lyapunov exponent is a mathematical tool used to represent the dynamics of chaotic systems. It is used to represent a measure of the separation rate of neighbouring trajectories in phase space. If a particular Lyapunov exponent is positive, it indicates that the system state will disperse rapidly in time and exhibit chaotic behaviour. In this case, a larger Lyapunov exponent indicates that the system has a more chaotic behaviour and the trajectories exhibit greater complexity and unpredictability. If it is negative, it indicates that the system state will tend to be stable.

The Lyapunov exponent of the system is $(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6) = (2.172, 1.445, 0.1488, 0.035, -0.064, -3.520)$ when the chaotic initial values are taken as (0.1, 0.2, 0.3, 0.4, 0.5, 0.6) respectively. The Lyapunov exponents are shown in Figure 5. It can be seen that the system has four positive Lyapunov exponents. This six-dimensional chaotic system possesses a more complex dynamic behaviour and the system is more stochastic. According to the determination criterion of a hyperchaotic system, the system has two Lyapunov exponents

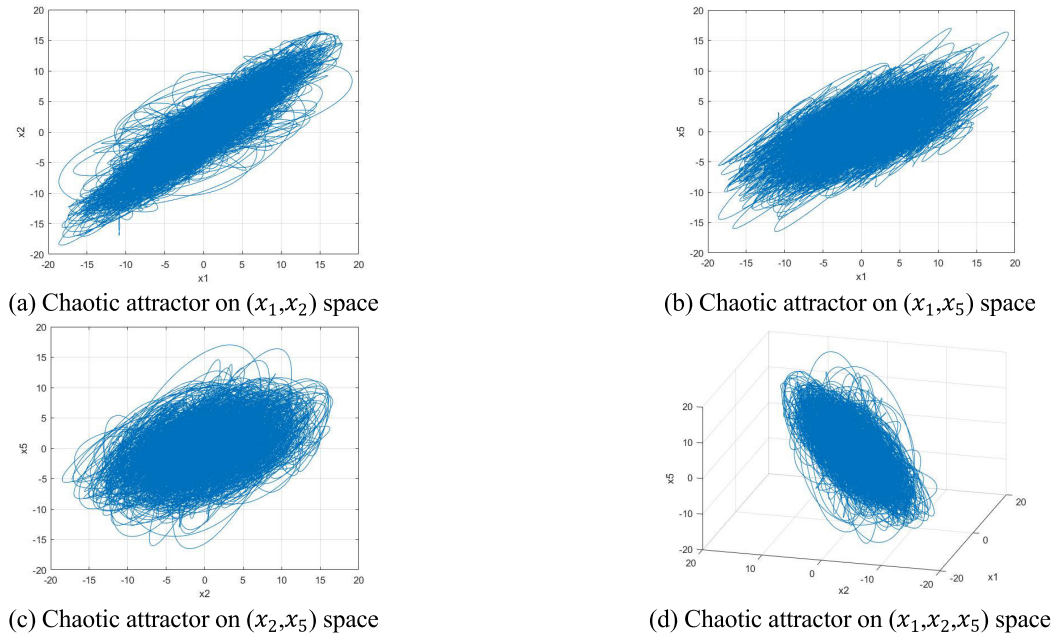


FIGURE 7. Attractor phase diagram of the chaotic system.

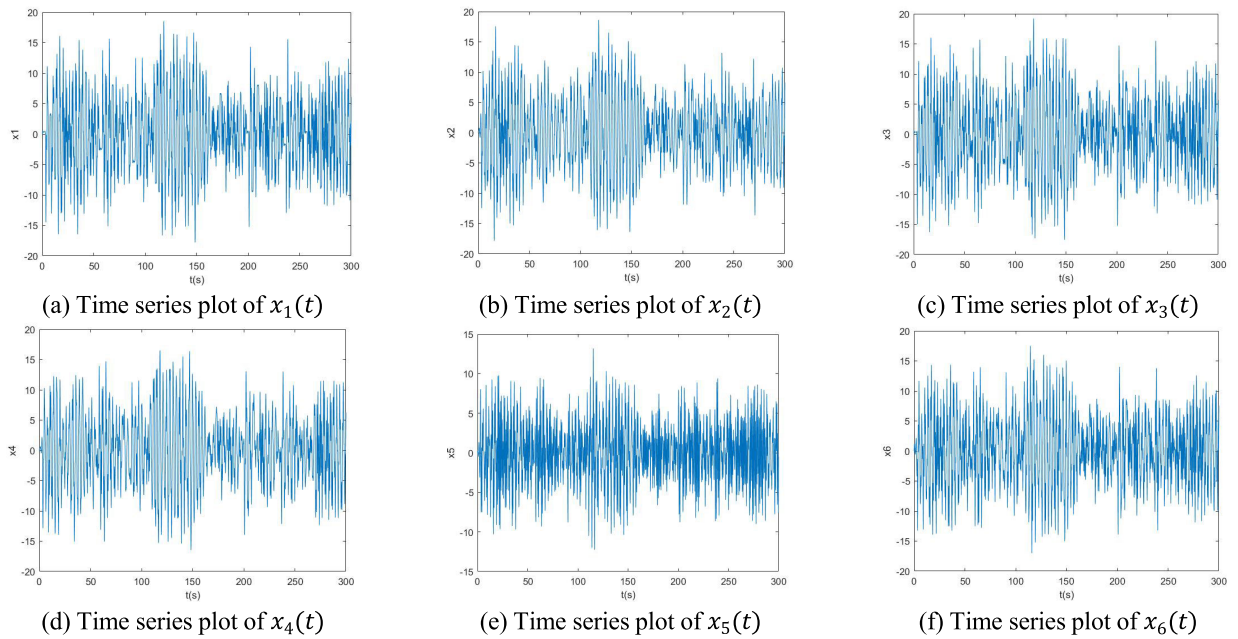


FIGURE 8. Six time series plots of chaotic systems.

greater than 0, so the system is hyperchaotic. The relevant formula is shown in Eq. (16).

$$\lambda_i = \lim_{t \rightarrow \infty} \lim_{\delta_0 \rightarrow 0} \frac{1}{t} \ln \left(\frac{\|\delta(t)\|}{\|\delta_0\|} \right) \quad (16)$$

where λ_i denotes the i -th Lyapunov exponent; t denotes time; $\delta(t)$ denotes the Euclidean distance between two state vectors at time t ; and δ_0 denotes the initial state vector $x(0)$ tiny perturbation.

The structure of chaotic systems is different from that of conventional systems in that they exhibit fractal properties, and the fractal dimension of the system can be quantified based on the Lyapunov dimension. This study uses the six Lyapunov exponents obtained to calculate the Lyapunov dimension of the chaotic system as 6.061 according to Eq. (17), where LE_i denotes the Lyapunov exponent of the i -th dimension, d_L denotes the dimension of the Lyapunov exponent, and j denotes the total number of dimensions of the chaotic system minus 1. From the above analysis, the

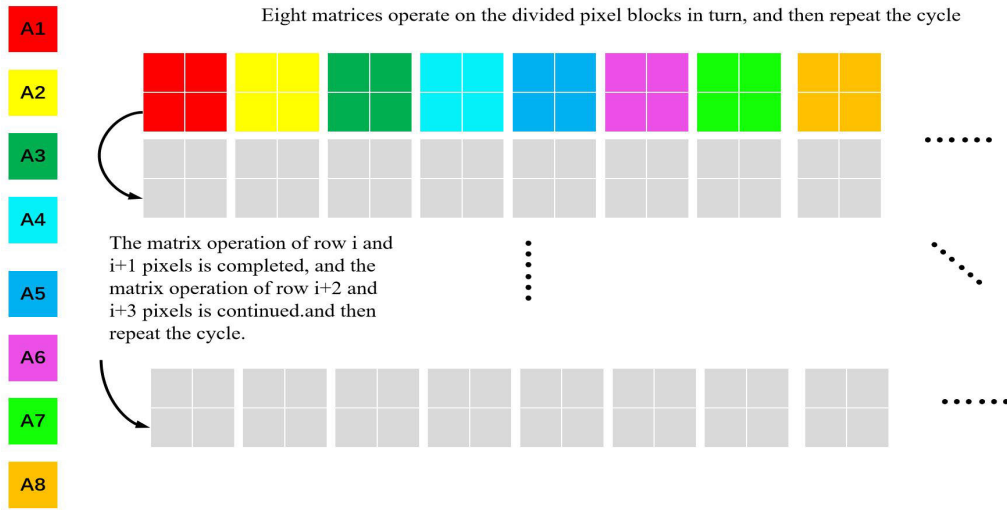


FIGURE 9. Operational process of image diffusion operation in the second encryption phase.

proposed six-dimensional chaotic system has four positive Lyapunov exponents, and the dimension of the Lyapunov exponent is a fraction that It shows that the system is chaotic.

$$d_L = j + \frac{1}{|L_{j+1}|} \sum_{i=1}^j LE_i = 5 + \frac{\sum_{i=1}^5 LE_i}{|LE_6|} \quad (17)$$

3) POINCARÉ CROSS SECTION

The Poincaré cross section is a mathematical construct widely used in the study of chaotic systems to capture and characterize chaotic trajectories in the state space of a system. The core idea of the Poincaré cross section is to project the state of a multidimensional chaotic system onto a low-dimensional subspace, thus transforming a high-dimensional complex dynamical problem into a low-dimensional one that is easier to analyze and understand. The motion characteristics of the intercept point can be observed from a lower dimension by observing the Poincaré cross section, the same Matlab R2021a is used to draw the Poincaré cross section of the system, the Poincaré cross section in the $x_1 - x_2$ plane and the $x_2 - x_5$ plane is shown in Figure 6. From the cross section, it can be seen that the intercept point forms a dense point with a hierarchical structure, thus verifying that the motion of the intercept point is in a chaotic state, and also verifying that the pseudo-random sequence generated by the chaotic system possesses randomness and unpredictability.

4) PHASE DIAGRAM AND TIMING DIAGRAM OF CHAOTIC ATTRACTORS FOR CHAOTIC SYSTEMS

From the above given system parameters, Matlab R2021a is used to plot the phase diagram and timing diagram of the system attractor, which is shown in Figure 7. From the phase diagram, it can be seen that the orbital trajectories are disordered and in stretching, showing chaotic characteristics.

From the timing diagram, it can be seen that the sequence presented is random in nature. The timing diagram of the system is shown in Figure 8.

In this subsection, a comprehensive analysis of the chaotic system is carried out. By analyzing the dissipation of the chaotic system, the rate of change of the phase space $\nabla \cdot V < 0$, indicates that the chaotic system is dissipative. Further, the chaotic system possesses four positive Lyapunov exponents and the Lyapunov dimension is fractional, which verifies the chaos of the system. The orbital trajectories show clear disorder in the Poincaré cross section and the phase diagram of the chaotic attractor, while the time series diagram shows the non-periodicity and irregularity of the system sequence. Based on the above quantitative and qualitative results, it can be analyzed that the designed chaotic system has chaos property.

C. APPLICATION OF THE FIBONACCI Q-MATRIX

The Fibonacci Q-matrix usually refers to a specific 2×2 matrix associated with the Fibonacci series, and the n -th power of the Fibonacci Q-matrix is defined by the Eq. (18). Where F_n is the n -th term of the Fibonacci series. This matrix demonstrates a nice property of the Fibonacci series in matrix form, where any term of the Fibonacci series can be computed efficiently by matrix multiplication.

The determinant of the Fibonacci Q-matrix can be expressed as Eq. (18). It is shown through Eq. (19) that the determinant of the Fibonacci Q-matrix is related to the parity of n in value. In the application of image encryption, the Fibonacci Q-matrix can be used in the diffusion phase of the second encryption process of the encryption algorithm.

$$Q^n = \begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix} \quad (18)$$

$$\text{Det}(Q^n) = F_{n+1}F_{n-1} - F_n^2 = (-1)^n \quad (19)$$

In this study, our object is a greyscale image of 256×256 pixel size. Since 256 is a number that can be divisible by more than one number, this study decomposes the image after the first scrambling operation in the second encryption process into multiple 2×2 matrices in a way that facilitates the execution of the diffusion operation that is crucial in this phase. Specifically, this study selected eight 2×2 matrices for our experiments, and the eight matrices are titled A1 to A8. These matrices serve not only as the basic units of the diffusion operation, but also as the key parameters in the encryption process. The eight matrices specifically selected:

$$\begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 3 & 1 \\ 4 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}, \\ \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}, \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix}$$

This study achieves image diffusion by performing matrix operations sequentially with these 8 matrices and the matrix obtained by image chunking in a certain order (first by columns and subsequently by rows). This process not only increases the security and complexity of the encryption process, but also further reduces the recognisability of the image content.

The specific process of the operational procedure and the formula for the diffusion operation are shown in Figure 9 and Eq. (20), respectively. Where E_2 denotes the image obtained from the first scrambling operation, E_3 denotes the image obtained after the diffusion operation, and H denotes the small matrix conforming to the properties of the Fibonacci Q-matrix. i and j denote the pixel positions, which take values in the ranges $i=1:3:5 \dots M$ and $j=1:3:5 \dots N$. We obtained the encrypted image after diffusion operation by performing cyclic matrix product operation on these matrices. This diffusion operation can further improve the security of image information.

$$\begin{bmatrix} E_{3i,j} & E_{3i,j+1} \\ E_{3i+1,j} & E_{3i+1,j+1} \end{bmatrix} \\ = \begin{bmatrix} E_{2i,j} & E_{2i,j+1} \\ E_{2i+1,j} & E_{2i+1,j+1} \end{bmatrix} \\ \begin{bmatrix} H_{i,j} & H_{i,j+1} \\ H_{i+1,j} & H_{i+1,j+1} \end{bmatrix} \bmod 256 \quad (20)$$

In the proposed second encryption process, the Knuth-Durstenfeld Shuffle algorithm is used to implement the second scrambling operation on the ciphertext image after the diffusion operation to obtain the final ciphertext image. This study uses the Knuth-Durstenfeld Shuffle algorithm to achieve the purpose of the scrambling operation by exchanging the pixel positions in the image to ensure that each element has an equal probability of appearing in any position. In this study, the pixel sum of the plaintext image is normalized to between 0-255 as the random number seed for the Knuth-Durstenfeld Shuffle algorithm, as shown in the Eq. (21). Where S_{min} denotes the smallest pixel value in the plaintext image, S_{max} denotes the largest pixel value in

the plaintext image, S denotes the pixel sum of the plaintext image, and S_{random} denotes the generated random seed.

$$S_{random} = \frac{S - S_{min}}{S_{max} - S_{min}} \times 255 \quad (21)$$

IV. EVALUATION

A. OVERVIEW

This study proposes a novel image double encryption and decryption method. In the first encryption process: This study proposes an improved generative adversarial network. We train the network by channel merging plaintext images and random greyscale images with Gaussian distribution characteristics as inputs; The generator and discriminator confront each other during training; after training is completed, the first encryption network and its corresponding decryption network are formed. The corresponding encryption and decryption network structure and network parameters are shown in Table 1 and Table 2. The network performs three downsampling operations to reduce the spatial dimension of the feature map. Subsequently, these downsampled feature maps were sent to nine sets of multi-level attention enhanced residual network structures designed for feature extraction and enhancement. On this basis, the feature map obtained after processing by the MAERS is directed to the FDFM, and finally an upsampling operation is performed to recover the original resolution of the feature map.

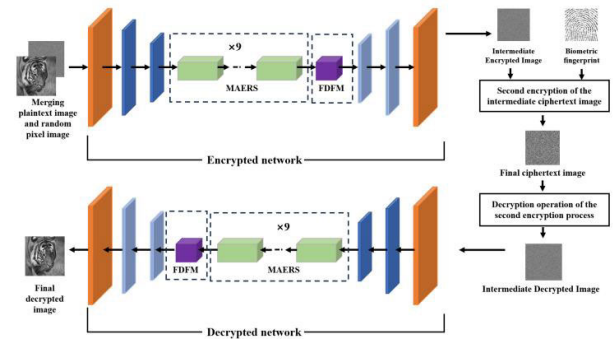


FIGURE 10. Image encryption and decryption process.

In the second encryption process: This study constructs a six-dimensional hyperchaotic system. With six complex key sequences generated by this system, three of the sequences are selected for collocation, and the merged sequence performs a scrambling operation on the intermediate ciphertext image obtained from the first encryption; Selecting eight 2×2 size matrices that conform to the Fibonacci Q-matrix property and perform diffusion operations on the images in order of first row then column; The normalized value of the pixel sum of the plaintext image is used as a random seed to further scramble the image using the Knuth-Durstenfeld Shuffle algorithm. With this, the encryption process is completed. The decryption process is the reverse of the encryption operation: The encrypted image is inversely scrambled by the

TABLE 1. Models and parameters of the encryption network.

Convolution layer name	Number	Convolution kernel	Input channels	Output channels	Parameters
Convolution1	1	7	2	64	6336
Convolution1	1	3	64	128	73856
Convolution1	1	3	128	256	294912
MAERS	9	3	256	256	5329152
FDFM+Deconvolution1	1	3	256	128	295040
Deconvolution1	1	3	128	64	73728
Deconvolution1	1	7	64	256	3137

TABLE 2. Models and parameters of the decryption network.

Convolution layer name	Number	Convolution kernel	Input channels	Output channels	Parameters
Convolution1	1	7	2	64	3180
Convolution2	1	3	64	128	73856
Convolution3	1	3	128	256	294912
MAERS	9	3	256	256	5329152
FDFM+Deconvolution1	1	3	256	128	295040
Deconvolution2	1	3	128	64	73728
Deconvolution3	1	7	64	1	3137

Knuth-Durstenfeld Shuffle algorithm; the encrypted image is scrambled by the Knuth-Durstenfeld Shuffle algorithm in reverse; the inverse matrix of the Fibonacci Q-matrix is used to perform the inverse operation of the diffusion operation on the encrypted image and the pseudo-random sequence used for the first scrambling operation of the second encryption is used. At this point, we complete the first decryption of the image; The ciphertext image after the first decryption process is decrypted by the decryption network obtained by training and the plaintext image is recovered. Figure 10 illustrates the image encryption and decryption process proposed in this study.

B. DATASETS AND EXPERIMENTAL ENVIRONMENT

In the testing and simulation experiments, the Div2K public dataset is used as a training set to train the network, and the performance of the proposed encryption and decryption algorithm is tested using classic image processing images such as Peppers, Boat and the POSTDAM remote sensing image public dataset as a test set, which is chosen as the initial value of the chaotic system for generating the fingerprint image from the standard fingerprint image library of FV2002. In addition, the proposed algorithm is tested in comparison with existing algorithms. In the comparison experiments, differential attack, correlation coefficient analysis, histogram statistical analysis, information entropy, key sensitivity and robustness analysis are used for the evaluation. For this experiment, this study used Pycharm and MATLAB 2021Ra, executing it on a server equipped with an Intel(R) Xeon(R) CPU E5-2620 v4 @ 2.10GHz and an NVIDIA GeForce GTX 1080 Ti GPU, as well as on a laptop with an Intel(R) Core(TM) i7-8750H CPU @ 2.20GHz and an NVIDIA GeForce GTX 1050 Ti GPU.

C. IMAGE ENCRYPTION AND DECRYPTION EFFECT ANALYSIS

This study evaluates the decryption effect using the Peppers image, Camerman image and Potsdam remote sensing image dataset. Figure 11 presents the original, encrypted and decrypted images of the experiments, it can be seen that the encrypted image's readable content has disappeared while the decrypted image is very close to the original image. In order to evaluate the performance of the image encryption and decryption algorithms in multiple ways, quantitative evaluation is performed using three metrics, namely, Mean Square Error (MSE), Peak Signal Noise Ratio (PSNR), and Structural Similarity Index (SSIM), which can help to identify the possible loss of image quality that may be introduced by the encryption process.

The mean square error (MSE) is a basic method for evaluating image quality, which calculates the average of the sum of squares of the pixel intensity differences between the original image and the processed image. The formula for MSE is shown in Eq. (22). Where $X(i, j)$ and $Y(i, j)$, are the pixel values of the original and processed images at position (i, j) respectively, M and N are the height and width of the image. Ideally, for a perfect encryption-decryption process, the MSE should be close to 0.

$$MSE = \frac{1}{M \times N} \sum_{i=1}^M \sum_{j=1}^N [X(i, j) - Y(i, j)]^2 \quad (22)$$

Peak Signal to Noise Ratio (PSNR) is commonly used in image encryption to measure the visual quality of the encrypted and decrypted image. PSNR is based on the MSE, which takes into account the possible maximum value of the pixels of the image, and the PSNR formula is shown in Eq. (23). Where MAX_x is the maximum possible pixel value of the image. For an 8-bit image, $MAX_x = 255$. A higher

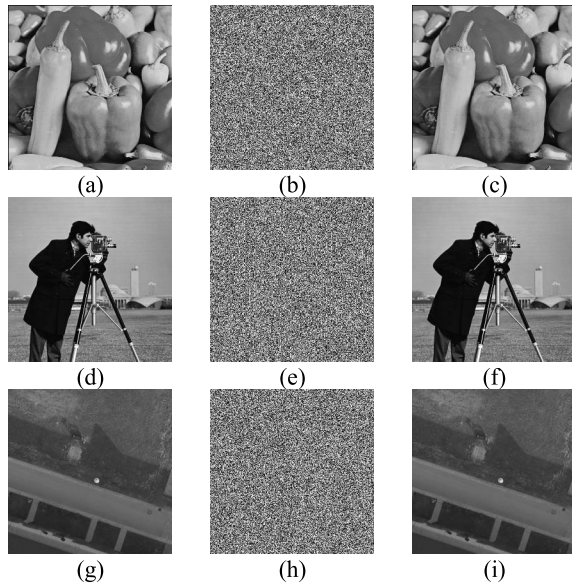


FIGURE 11. Image encryption and decryption effect diagrams: (a) original Peppers image; (b) encryption effect diagram of the Peppers image; (c) decryption effect diagram of the Peppers image; (d) original Cameraman image; (e) encryption effect diagram of the Cameraman image; (f) decryption effect diagram of the Cameraman image; (g) original image of the remote sensing image; (h) remote sensing image encryption effect; (i) remote sensing image decryption effect.

TABLE 3. Experimental results of PSNR and SSIM obtained for images in different experimental methods.

Method	PSNR	SSIM
Proposed(Peppers)	33.8483	0.9337
[43](Peppers)	33.8756	0.9329
[44](Peppers)	32.9624	0.9374
[45](Peppers)	33.5869	0.9267
[46](Peppers)	32.7833	0.9289
[47](Peppers)	33.8056	0.9314
[48](Peppers)	33.8419	0.9325
Proposed(Cameraman)	33.3145	0.9383
[43](Cameraman)	33.5871	0.9358
[44](Cameraman)	33.3094	0.9298
[45](Cameraman)	32.5876	0.9217
[46](Cameraman)	33.2384	0.9306
[47](Cameraman)	33.3375	0.9374
[48](Cameraman)	33.3042	0.9287
Proposed(POTSDAM)	35.6531	0.9412
[43](POTSDAM)	35.6014	0.9368
[44](POTSDAM)	33.7502	0.9354
[45](POTSDAM)	32.9485	0.9264
[46](POTSDAM)	32.8946	0.9255
[47](POTSDAM)	35.5914	0.9295
[48](POTSDAM)	35.6215	0.9330

PSNR indicates better image quality and less visual distortion introduced by the encryption and decryption process.

$$PSNR = 20 \cdot \log_{10} \left(\frac{MAX_x}{\sqrt{MSE}} \right) \quad (23)$$

The structural similarity index (SSIM) provides a more detailed method of evaluating image quality, which assesses

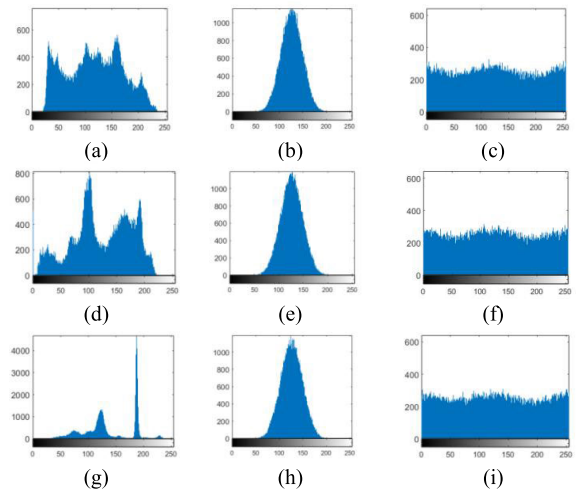


FIGURE 12. Histogram analysis: (a) Histogram of the original Barbara graph; (b) Histogram of the middle ciphertext of the Barbara graph; (c) Histogram of the final ciphertext of the Barbara graph; (d) Histogram of the original Peppers graph; (e) Histogram of the middle ciphertext of the Peppers graph; (f) Histogram of the final ciphertext of the Peppers graph; (g) Histogram of the original house graph; (h) Histogram of the final ciphertext of the house graph; (i) House graph final ciphertext histogram.

the similarity of an image in terms of brightness, contrast, and structure. The SSIM formula is shown in Eq. (24). SSIM takes values ranging from -1 to 1 , where 1 indicates that the two images are identical. High values of SSIM indicate high visual similarity between the images. Where μ_x and μ_y represent the mean luminance of images x and y respectively, σ_x^2 and σ_y^2 are the variance of images x and y respectively, and σ_{xy} is the covariance of x and y . C_1 and C_2 are small constants introduced to avoid the denominator being zero. Table 3 gives the experimental results of PSNR and SSIM for the original and decrypted images of Peppers image, Cameraman image and Potsdam remote sensing image and compared with different experimental methods.

The experimental results show that the encryption and decryption method proposed in this study is able to complete the image decryption operation well, especially for remote sensing image datasets. It is able to maintain a good balance between image quality and structure fidelity.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (24)$$

D. HISTOGRAM STATISTICAL ANALYSIS

The histogram is a fundamental tool in image analysis that shows the distribution of pixel values in an image. In the context of image encryption, histogram analysis is used to evaluate whether an encryption algorithm is effective in randomizing the distribution of pixel values in an image, thus preventing leakage of image content through statistical analysis. The histogram of the encrypted image should be close to a uniform distribution, meaning that every possible pixel value has the same probability of occurrence, making it impossible

for an attacker to obtain any useful information from the histogram. The experimental results are shown in Figure 12.

The pixel distribution of the intermediate ciphertext image conforms to the Gaussian distribution distribution, and the histogram of the final encrypted image obeys a uniform distribution. The attacker has been unable to get the effective information of the plaintext image from it.

E. CORRELATION COEFFICIENT ANALYSIS

In the field of digital image processing, analyzing the correlation between elements of two or more variables is a way to measure how closely these variables are connected. In image data, neighboring pixels usually exhibit significant correlations between them, leading to the possibility that one pixel may reveal information about its neighboring pixels. This information leakage phenomenon allows a potential attacker to exploit this property by inferentially predicting the next pixel grayscale values, which in turn may recover the original plaintext image. In digital images, a high degree of correlation exists between neighboring pixels as they usually exhibit similar intensity values to each other. To defend against statistical attacks, these high inter-pixel correlations must be effectively destroyed. In this paper, 10,000 pairs of neighboring pixels in the vertical, horizontal and diagonal directions are randomly selected from both normal and cryptographic images and the correlation coefficients of the two neighboring pixels are computed using the Eq. (25). Where N is the number of pixel pairs, x, y are the pixel values at neighboring locations, $E(x)$ denotes the mean, $D(x)$ is the variance and $cov(x, y)$ is the covariance.

$$\left\{ \begin{array}{l} r_{xy} = \frac{cov(x, y)}{\sqrt{D(x)}\sqrt{D(y)}} \\ cov(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - E(x))(y_i - E(y)) \\ D(x) = \frac{1}{N} \sum_{i=1}^N (x_i - E(x))^2 \\ E(x) = \frac{1}{N} \sum_{i=1}^N x_i \end{array} \right. \quad (25)$$

Figure 13 depicts the neighboring pixel correlations in the three directions (horizontal, vertical and diagonal) of the pepper picture. From the figure it can be known that the proposed encryption method makes the pixels of the encrypted image uniformly distributed and effectively eliminates the correlation of neighboring pixels in the three directions. The closer the value of the neighboring pixel correlation is to 1, the stronger the correlation between neighboring pixels, and the closer the value of neighboring pixel correlation is to 0, the weaker the correlation between neighboring pixels, as shown in the Table 4, the value of correlation between neighboring pixels of Peppers and Cameraman images is evaluated, and it can be found that the coefficients of the three directions are close to 0, which effectively destroys correlation between neighboring pixels and decreases the readability of the image

content. A correlation comparison test is also performed, which shows that we have improved the effectiveness of destroying the adjacent pixel correlation in both horizontal and vertical directions.

F. KEY SPACE ANALYSIS

In image encryption, the size and characteristics of the key space are critical to ensure encryption security. Having a key space large enough to make brute force cracking impractical, the size of the key space is usually recommended to be at least $> 2^{100}$ [48]. The environment used in this experiment is in a 64-bit architecture where the basic unit of data processing is a 64-bit binary number. The encryption key for the pro-posed method includes the encryption network parameters and the biometric fingerprint image. The encryption key for the first encryption process involves the encryption network parameters, the total sum of the convolutional layer parameters is 6076161, and the range of network parameter values is in $[-1, 1]$. In the first disruption phase of the second encryption process, the biometric fingerprint image, on the other hand, consists of an eight-bit binary representation of the grayscale map with a parameter sum of $256 \times 256 \times 1 = 65536$. The key for the diffusion phase of the second encryption process consists of eight 2×2 matrices with a parameter sum of $(2^{64})^{32}$. In summary, the encryption process key space sums to $(2^{64})^{6076161} + (2^8)^{65536} + (2^{64})^{32}$. Similarly, the decryption key includes the decryption network parameters and the biometric fingerprint image, and the decryption network parameters in the first decryption process sum up to 1335981, and the second decryption process has the same key space as the second encryption process, so that the decryption key space sums up to $(2^{64})^{6073005} + (2^8)^{65536} + (2^{64})^{32}$ and the total key space size is $(2^{64})^{6076161} + (2^{64})^{6073005} + (2^8)^{65537} + (2^{64})^{33} \approx 2^{388874304} \gg 2^{100}$. The key space satisfies the higher security level requirements and is sufficient to resist brute force attacks. In addition, the key space of the related experiments is also calculated, and it can be seen from the Table 5 that the key space of the proposed method is larger and more effective to resist brute force attacks in terms of key space.

G. INFORMATION ENTROPY

The concept of information entropy was originally introduced by Claude Shannon in information theory, and this metric is usually expressed as the average amount of information present in an image or message. For an image, information entropy can be defined as a measure of uncertainty in the distribution of pixel values in the image. Specifically, the information entropy $H(X)$ can be calculated by the Eq. (26).

$$H(X) = \sum_{i=0}^{2^L-1} p(x_i) \log_2 \frac{1}{p(x_i)} \quad (26)$$

where L represents the length of the image pixel and $p(x_i)$ represents the probability of x_i occurrence. The closer the value of the theoretical information entropy is to 8 bits, the more chaotic the pixel distribution is represented. The information

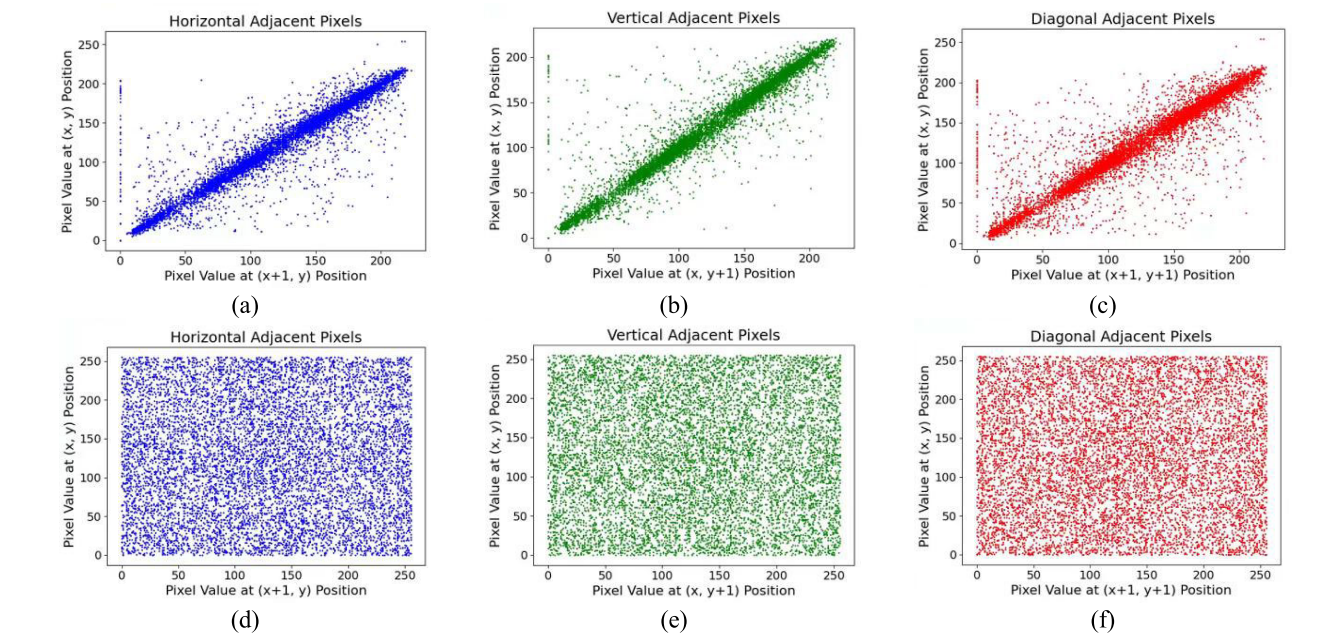


FIGURE 13. Peppers adjacent pixel correlation analysis:(a) Adjacent pixel correlation graph in the horizontal direction of the original image; (b) Correlation diagram of adjacent pixels in the vertical direction of the original image; (c) Correlation diagram of adjacent pixels in the diagonal direction of the original image; (d) Correlation graph of adjacent pixels in the horizontal direction of the encrypted image; (e) Correlation graph of adjacent pixels in the vertical direction of the encrypted image; (f) Correlation graph of adjacent pixels in the diagonal direction of the encrypted image.

TABLE 4. Correlation coefficients of two neighboring pixels in an image obtained by different algorithms in horizontal, vertical and diagonal directions.

Graphics	Direction	Plaintext image	[43]	[46]	[45]	[44]	[47]	[48]	Proposed
Peppers	Horizontal	0.9488	0.0003	0.0198	-0.0156	0.0038	0.0084	0.0029	0.0005
	Vertical	0.9547	0.0031	0.0233	0.0242	0.0059	-0.0073	0.0014	0.0023
	Diagonal	0.9140	0.0016	0.0341	-0.0278	0.0064	0.0052	0.0037	-0.0048
Cameraman	Horizontal	0.9365	0.0010	0.0171	0.0167	-0.0027	-0.0096	-0.0036	-0.0001
	Vertical	0.9624	0.0019	-0.0210	-0.0146	-0.0036	-0.0147	0.0066	0.0004
	Diagonal	0.9110	0.0034	0.0247	-0.0196	0.0130	0.0025	0.0029	0.0132

TABLE 5. Comparison of key space between different experiments.

Proposed	[43]	[44]	[45]	[46]	[47]	[48]
$2^{388874304}$	2^{280}	10^{315} $\approx 2^{1046}$	$(2^8)^{196608} + 2^{128}$ $\approx 2^{1572864}$	$2^{176750976}$	2^{509}	2^{260}

TABLE 6. Image information entropy test.

Image	Plaintext image	Ciphertext image	[43]	[44]	[45]	[46]	[47]	[48]
Peppers	7.5327	7.9990	7.9986	7.9988	7.9981	7.9976	7.9977	7.9990
Cameraman	7.0097	7.9984	7.9977	7.9979	7.9976	7.9974	7.9973	7.9982

entropy obtained by different methods is calculated through comparative tests and the results are shown in Table 6. As can be seen from Table 6, the image entropy values of our method on both Peppers and Cameraman images are close to 8 bits and are higher with the compared methods. This means that the method proposed in this study makes the cipher image

well randomized and the information leakage during the encryption process is negligible.

H. DIFFERENTIAL ATTACK

Differential attack is an analytical technique for image encryption algorithms, which uses the differences in the

encryption algorithm to attempt to decipher the input image. This attack method is particularly suitable for image encryption because image data usually contains highly redundant information and specific statistical features. In a differential attack, two images with only minor differences in very few pixels are usually prepared as inputs, and then the two images are encrypted separately and the encrypted results are compared and analyzed. By observing closely the differences between the two encrypted images, an attacker can gain important information about the way the encryption algorithm handles subtle changes. If an encryption algorithm is not sensitive enough to small changes in the image, these small changes may reveal breaks in the encrypted image, thus revealing information about the encryption process. Thus, differential attacks can reveal potential weaknesses in the encryption algorithm.

In order to measure the resistance of image encryption algorithms to differential attacks, two key metrics are usually used: the number of pixel change rate (NPCR) and the uniform average change intensity (UACI), and the related formulas are shown in Eq. (27) and Eq. (28). NPCR measures the proportion of pixel changes in an encrypted image due to a single pixel change, reflecting the algorithm's sensitivity to small changes. UACI quantifies the image quality changes by comparing the difference between the pixel values of the plaintext image and the ciphertext image, reflecting the overall degree of change in the encrypted image. These two metrics were used to quantify the difference between the two ciphertext images, and their corresponding ideal values were 99.61% and 33.46%, respectively. The closer the NPCR and UACI are to the ideal values, the more sensitive the encryption algorithm is to the security key.

$$NPCR = \frac{\sum_{i,j} D(i,j)}{W \times H} \times 100 \quad (27)$$

$$UACI = \frac{1}{W \times H} \left[\sum_{i,j} \frac{|c_1(i,j) - c_2(i,j)|}{255} \right] \times 100 \quad (28)$$

where W and H are the width and height of the image, respectively, and $C_1(i,j)$ and $C_2(i,j)$ are the images before and after the pixel value is changed on the plaintext image position (i,j) .

By changing the pixel values of the plaintext image, it can be seen from the Table 7 that a small change in the plaintext image leads to a change in the ciphertext image. The proposed encryption algorithm corresponds to NPCR and UACI with better results, whereas the experimental results of Peppers image are more obvious. The experimental analysis shows that the proposed encryption method is effective against differential attacks.

I. SELECT SPECIFIC PLAINTEXT ATTACK ANALYSIS

The Choice Specific Plaintext Attack (CPA) is an important method of analysis when studying the security of image encryption algorithms. This method is particularly suitable for testing the algorithm's ability to handle extreme

TABLE 7. By comparing the experimentally obtained NPCR and UACI.

Image	NPCR(%)	UACI(%)
Ours(streamandbridge1)	99.58	33.62
Ours(mandrill)	99.60	33.52
Ours(Peppers)	99.65	33.89
[47](Peppers)	99.61	33.47
[48](Peppers)	99.60	33.48
[43](Peppers)	99.61	33.74
[44](Peppers)	99.57	33.36
[45](Peppers)	99.50	33.23
[46](Peppers)	99.43	32.80

situations, where the analysis of encryption of all-black and all-white images is particularly important. The all-black image (all pixel values are 0) and the all-white image (all pixel values are maximum brightness values) represent the two extreme cases possible in image data. In this analysis, the selection of all-black and all-white images is not random but based on their extreme consistency in pixel distribution, which provides a unique challenge in evaluating cryptographic algorithms. Ideally, a strong image encryption algorithm should be able to transform these extreme cases into seemingly random, unrecognizable encrypted images, preventing an attacker from recovering any meaningful information from the encrypted images. The encryption of all-black and all-white images is shown in Figure 14.

The results of selecting a particular plaintext attack to analyze are crucial for evaluating the security of the encryption algorithm. If the encrypted all-black or all-white images show recognizable patterns or statistical regularities, this may indicate potential security weaknesses in the encryption algorithm. Therefore, this analysis not only helps to reveal possible information leakage points in the encryption process, but also emphasizes the need to design encryption algorithms that ensure good encryption performance for all possible plaintext inputs.

J. KEY SENSITIVITY ANALYSIS

Key sensitivity is the sensitivity of the encryption algorithm to the change of key, which directly reflects the security of the encryption process. An image encryption algorithm with high key sensitivity ensures that even the slightest change in the key will result in a fundamental difference in the encrypted output, making it impossible for an attacker to crack the encrypted image by guessing or fine-tuning the key.

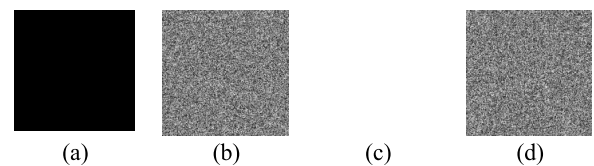


FIGURE 14. All-black and all-white images and their final encrypted images: (a) All-black image; (b) Encrypted image of all-black image; (c) All-white image; (d) Encrypted image of all-white image.

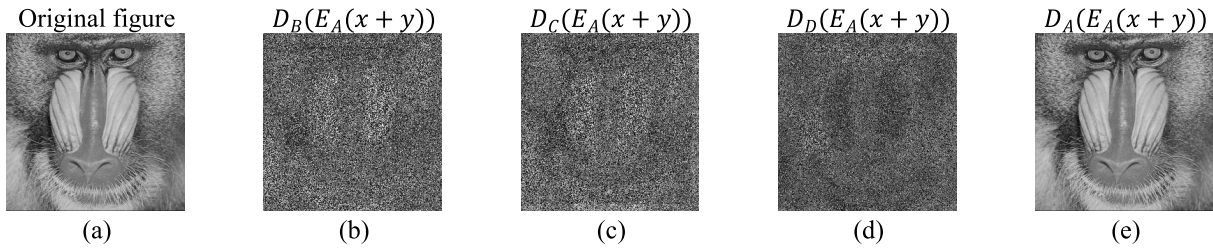


FIGURE 15. Decrypted image obtained by training four network structures.

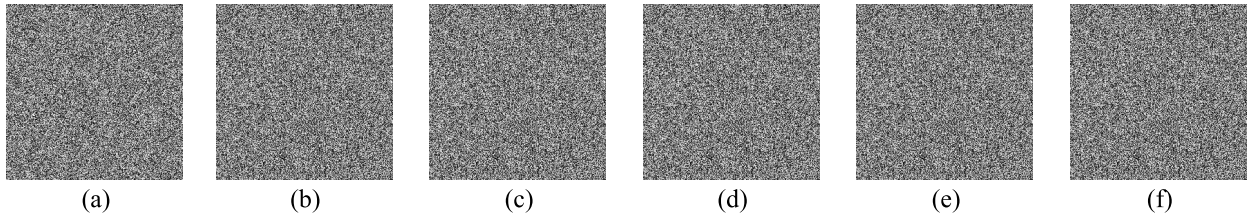


FIGURE 16. The decrypted image obtained by changing the initial value of the chaotic system, the amount of change is: (a) change the initial value of the first dimension of the chaotic system; (b) change the initial value of the second dimension of the chaotic system; (c) change the initial value of the third dimension of the chaotic system; (d) change the initial value of the fourth dimension of the chaotic system; (e) change the initial value of the fifth dimension of the chaotic system; (f) change the initial value of the sixth dimension of the chaotic system.

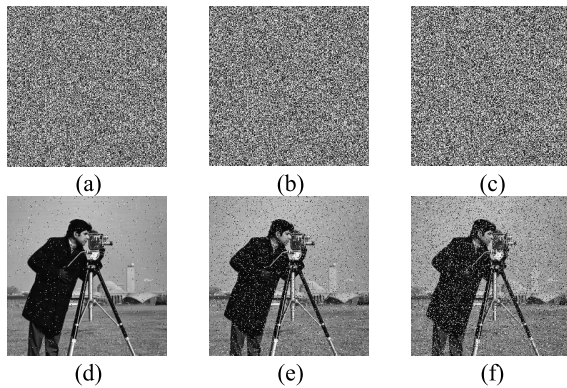


FIGURE 17. Salt-and-pepper noise and corresponding decryption images: (a) the ciphertext is subject to 1% salt-and-pepper noise; (b) the ciphertext is subject to 5% salt-and-pepper noise; (c) the ciphertext is subject to 10% salt-and-pepper noise; (d), (e), and (f) are the decryption images of (a), (b), and (c), respectively.

In the test, by setting the same parameters and network structure for the experiment, training four times to get four network structure A,B,C,D, respectively, corresponding to this encryption network E_A, E_B, E_C, E_D , and decryption network D_A, D_B, D_C, D_D . Firstly, the test set is used to encrypt through the encryption network E_A and then decrypted by D_A, D_B, D_C, D_D respectively and the decrypted image is shown in Figure 15. Where x denotes the plaintext image and y denotes the random pixel image combined with the plaintext x channel. The experimental results show that the encryption and decryption networks exist in sets, and even if the attacker obtains the structure of the network but not the trained parameters, he still cannot get the correct plaintext information.

In addition, we use a pair of encryption network E_A and decryption network D_A , in other words, we ensure that the

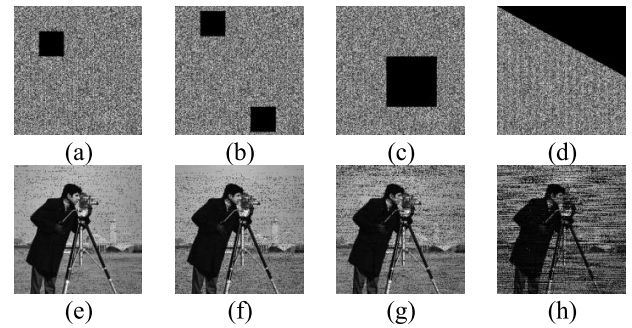


FIGURE 18. Cropping attack and restoration effect diagrams: (a) a 50×50 pixel loss; (b) two 50×50 pixel losses; (c) a 100×100 pixel loss; (d) a pixel loss in the upper triangular portion; (e), (f), (g), and (h) are decryption images of (a), (b), (c), and (d), respectively.

key of the first encryption and decryption process remains unchanged, and at the same time, the key in the second encryption is changed. For each of the six initial values of the chaotic system during the second encryption process add 10^{-16} . The results are shown in Figure 16. It can be noticed from the figure that it is no longer possible to restore the plaintext image back. Therefore, even though the attacker obtains the equations of the chaotic system, it is still unable to restore the plaintext image due to the strong sensitivity of the chaotic system to the initial value, which has a very small deviation. It is proved that the leakage of the chaotic system in this study does not threaten the security of the key.

K. ROBUSTNESS ANALYSIS

In practice, encrypted images may suffer from various interferences and attacks during transmission, such as noise, and distortion, degradation, and contamination caused by communication noise are common, and all of these factors will

have a certain impact on the decryption of the terminal image. A robust image encryption algorithm should be able to maintain the confidentiality of the encrypted information despite these unfavorable conditions, and be able to correctly recover the original image at the receiving end, or at least maintain the identifiability of the image content without complete loss of image information due to interference.

During the test, the effect of image encryption and decryption was demonstrated by using the cropping attack and salt-and-pepper noise attack, two security threats, which can significantly damage the integrity and usability of images. The salt-and-pepper attack reduces the visual quality of the image by introducing random white and black dot noise into the image. In this experiment, 1% salt-and-pepper noise, 5% salt-and-pepper noise and 10% salt-and-pepper noise are used to attack the ciphertext image respectively. The cropping attack targeted the destruction of important information by deleting specific areas of the image. In this experiment, one 50×50 area, one 100×100 area and two 50×50 areas were cropped to the ciphertext. These attacks may occur even in the case of image encryption. Therefore, adopting robust encryption algorithm and strengthening image integrity verification become the key strategies to ensure image security. The salt-and-pepper noise attack and cropping attack are shown in Figure 17 and Figure 18 respectively. According to the experimental results, the main readable information of the image can still be obtained if some information is lost or contaminated.

V. CONCLUSION

In this paper, an image double encryption and decryption method based on the combination of deep learning and traditional methods is proposed. The encryption method incorporates related techniques including improved generative adversarial networks, six-dimensional chaotic systems, Fibonacci Q-matrix and classical shuffling algorithms. In the proposed method, the image is encrypted for the first time by a trained encryption network. In the second encryption process, the biometric fingerprint is used to provide the initial value for the six-dimensional chaotic system, and three sequences are selected from the six pseudo-random sequences obtained by iteration for the first scrambling operation. Then, eight matrices conforming to Fibonacci Q-matrix properties are extracted for diffusion operations. Finally, the second scrambling operation is performed by the classical algorithm to obtain the final encrypted image. We perform qualitative and quantitative analysis by differential attack, correlation coefficient analysis, PSNR, SSIM, key space analysis, robustness analysis and other metrics. According to the comparison test, it is shown that the proposed method is better than other methods.

REFERENCES

- [1] M. A. Al-Khasawneh, W. Abu-Ulbeh, and A. M. Khasawneh, "Satellite images encryption review," in *Proc. Int. Conf. Intell. Comput. Hum.-Comput. Interact. (ICHCI)*, Dec. 2020, pp. 121–125.
- [2] N. Almalki and H. A. Alsuwat, "A systematic literature review on security challenges in image encryption algorithms for medical images," *Int. J. Comput. Sci. Netw. Secur.*, vol. 22, no. 6, pp. 75–82, 2022.
- [3] G. Ye and L. Guo, "A visual meaningful encryption and hiding algorithm for multiple images," *Nonlinear Dyn.*, vol. 112, no. 16, pp. 14593–14616, Aug. 2024.
- [4] G. Ye, M. Liu, W.-S. Yap, and B.-M. Goi, "Reversible image hiding algorithm based on compressive sensing and deep learning," *Nonlinear Dyn.*, vol. 111, no. 14, pp. 13535–13560, Jul. 2023.
- [5] N.-R. Zhou, L.-L. Hu, Z.-W. Huang, M.-M. Wang, and G.-S. Luo, "Novel multiple color images encryption and decryption scheme based on a bit-level extension algorithm," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 122052.
- [6] M. K. Khairullah, A. A. Alkahtani, M. Z. Bin Baharuddin, and A. M. Al-Jubari, "Designing 1D chaotic maps for fast chaotic image encryption," *Electronics*, vol. 10, no. 17, p. 2116, Aug. 2021.
- [7] S. Zhu, X. Deng, W. Zhang, and C. Zhu, "Secure image encryption scheme based on a new robust chaotic map and strong S-box," *Math. Comput. Simul.*, vol. 207, pp. 322–346, May 2023.
- [8] D. Ding, W. Wang, Z. Yang, Y. Hu, J. Wang, M. Wang, Y. Niu, and H. Zhu, "An n-dimensional modulo chaotic system with expected Lyapunov exponents and its application in image encryption," *Chaos, Solitons Fractals*, vol. 174, Sep. 2023, Art. no. 113841.
- [9] Q. Liang and C. Zhu, "A new one-dimensional chaotic map for image encryption scheme based on random DNA coding," *Opt. Laser Technol.*, vol. 160, May 2023, Art. no. 109033.
- [10] M. U. Rehman, "Quantum-enhanced chaotic image encryption: Strengthening digital data security with 1-D sine-based chaotic maps and quantum coding," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 36, no. 3, Mar. 2024, Art. no. 101980.
- [11] G. Ye, H. Wu, M. Liu, and Y. Shi, "Image encryption scheme based on blind signature and an improved Lorenz system," *Expert Syst. Appl.*, vol. 205, Nov. 2022, Art. no. 117709.
- [12] Z. Guo, S.-H. Chen, L. Zhou, and L.-H. Gong, "Optical image encryption and authentication scheme with computational ghost imaging," *Appl. Math. Model.*, vol. 131, pp. 49–66, Jul. 2024.
- [13] N.-R. Zhou, L.-J. Tong, and W.-P. Zou, "Multi-image encryption scheme with quaternion discrete fractional tchebyshev moment transform and cross-coupling operation," *Signal Process.*, vol. 211, Oct. 2023, Art. no. 109107.
- [14] M. Es-Sabry, N. El Akkad, M. Merras, K. Satori, W. El-Shafai, T. Altameem, and M. M. Fouda, "Securing images using high dimensional chaotic maps and DNA encoding techniques," *IEEE Access*, vol. 11, pp. 100856–100878, 2023.
- [15] J. Wang, L. Liu, M. Xu, and X. Li, "A novel content-selected image encryption algorithm based on the LS chaotic model," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 34, no. 10, pp. 8245–8259, Nov. 2022.
- [16] H. Huang and Z. Cai, "Duple color image encryption system based on 3-D nonequilateral Arnold transform for IIOT," *IEEE Trans. Ind. Informat.*, vol. 19, no. 7, pp. 8285–8294, Jul. 2023.
- [17] X.-D. Liu, Q.-H. Chen, R.-S. Zhao, G.-Z. Liu, S. Guan, L.-L. Wu, and X.-K. Fan, "Quantum image encryption algorithm based on four-dimensional chaos," *Frontiers Phys.*, vol. 12, Mar. 2024, Art. no. 1230294.
- [18] T. Li, W. Fan, J. Wu, and D. Zhang, "Image encryption based on a fractional-order hyperchaotic system and fast row-column-level joint permutation and diffusion," *Nonlinear Dyn.*, vol. 112, no. 12, pp. 10555–10581, Jun. 2024.
- [19] A. A. Neamah, "An image encryption scheme based on a seven-dimensional hyperchaotic system and Pascal's matrix," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 35, no. 3, pp. 238–248, Mar. 2023.
- [20] X. Ning, Q. Dong, S. Zhou, Q. Zhang, and N. K. Kasabov, "Construction of new 5D Hamiltonian conservative hyperchaotic system and its application in image encryption," *Nonlinear Dyn.*, vol. 111, no. 21, pp. 20425–20446, Nov. 2023.
- [21] S. Yan, L. Li, W. Zhao, and B. Gu, "Design of a new four-dimensional chaotic system and its application to color image encryption," *Nonlinear Dyn.*, vol. 111, no. 18, pp. 17519–17545, Sep. 2023.
- [22] J. Chen, L. Chen, and Y. Zhou, "Cryptanalysis of a DNA-based image encryption scheme," *Inf. Sci.*, vol. 520, pp. 130–141, May 2020.
- [23] X. Yao, R. Xie, X. Zan, Y. Su, P. Xu, and W. Liu, "A novel image encryption scheme for DNA storage systems based on DNA hybridization and gene mutation," *Interdiscipl. Sci., Comput. Life Sci.*, vol. 15, no. 3, pp. 419–432, Sep. 2023.

- [24] O. S. Faragallah, E. A. Naeem, H. S. El-sayed, and F. E. Abd El-Samie, "Efficient compression processing of optically DCT-based DRPE encrypted images," *Opt. Quantum Electron.*, vol. 54, no. 5, p. 273, May 2022.
- [25] J. Wang, M. Zhang, X. Tong, and Z. Wang, "An image compression encryption scheme based on chaos and SPECK-DCT hybrid coding," *Nonlinear Dyn.*, vol. 112, no. 11, pp. 9581–9602, Apr. 2024.
- [26] X. Wang, C. Liu, and D. Jiang, "A novel triple-image encryption and hiding algorithm based on chaos, compressive sensing and 3D DCT," *Inf. Sci.*, vol. 574, pp. 505–527, Oct. 2021.
- [27] S. K. Rao, N. K. Nishchal, and A. AlFalou, "Optical asymmetric image encryption using vectorial light field encoding," *Opt. Commun.*, vol. 554, Mar. 2024, Art. no. 130097.
- [28] X. Jiang, Y. Xie, B. Liu, J. Chai, Y. Ye, T. Song, M. Feng, and H. Yuan, "Image encryption based on actual chaotic mapping using optical reservoir computing," *Nonlinear Dyn.*, vol. 111, no. 16, pp. 15531–15555, Aug. 2023.
- [29] Y. Niu, H. Zhou, and X. Zhang, "Image encryption scheme based on improved four-dimensional chaotic system and evolutionary operators," *Sci. Rep.*, vol. 14, no. 1, p. 7033, Mar. 2024.
- [30] B. Long, Z. Chen, T. Liu, X. Wu, C. He, and L. Wang, "A novel medical image encryption scheme based on deep learning feature encoding and decoding," *IEEE Access*, vol. 12, pp. 38382–38398, 2024.
- [31] X. Chai, Y. Wang, X. Chen, Z. Gan, and Y. Zhang, "TPE-GAN: Thumbnail preserving encryption based on GAN with key," *IEEE Signal Process. Lett.*, vol. 29, pp. 972–976, 2022.
- [32] B. Wang and K.-T. Lo, "Autoencoder-based joint image compression and encryption," *J. Inf. Secur. Appl.*, vol. 80, Feb. 2024, Art. no. 103680.
- [33] Q. Zhou, X. Wang, M. Jin, L. Zhang, and B. Xu, "Optical image encryption based on two-channel detection and deep learning," *Opt. Lasers Eng.*, vol. 162, Mar. 2023, Art. no. 107415.
- [34] Q.-X. Huang, W. L. Yap, M.-Y. Chiu, and H.-M. Sun, "Privacy-preserving deep learning with learnable image encryption on medical images," *IEEE Access*, vol. 10, pp. 66345–66355, 2022.
- [35] Y. Ding, G. Wu, D. Chen, N. Zhang, L. Gong, M. Cao, and Z. Qin, "DeepEDN: A deep-learning-based image encryption and decryption network for Internet of Medical Things," *IEEE Internet Things J.*, vol. 8, no. 3, pp. 1504–1518, Feb. 2021.
- [36] Z. Liu and R. Xue, "Visual image encryption based on compressed sensing and cycle-GAN," *Vis. Comput.*, vol. 40, no. 8, pp. 5857–5870, Aug. 2024.
- [37] E. Abdellatef, E. A. Naeem, and F. E. A. El-Samie, "DeepEnc: Deep learning-based CT image encryption approach," *Multimedia Tools Appl.*, vol. 83, no. 4, pp. 11147–11167, Jan. 2024.
- [38] Y. Chen, H. An, and Z. Li, "The function cascade synchronization approach with uncertain parameters or not for hyperchaotic systems," *Appl. Math. Comput.*, vol. 197, no. 1, pp. 96–110, Mar. 2008.
- [39] X. Wu, Z.-H. Guan, and Z. Wu, "Adaptive synchronization between two different hyperchaotic systems," *Nonlinear Anal., Theory, Methods Appl.*, vol. 68, no. 5, pp. 1346–1351, Mar. 2008.
- [40] X. Wu, C. Zhong, B. Long, T. Liu, C. He, and L. Wang, "Chaotic image encryption algorithm based on dynamic hachimoji DNA coding and computing," *Phys. Scripta*, vol. 99, no. 4, Mar. 2024, Art. no. 045252.
- [41] M. Kopp and A. Kopp, "A new 6D chaotic generator: Computer modelling and circuit design," *Int. J. Eng. Technol. Innov.*, vol. 12, no. 4, pp. 288–307, Jun. 2022.
- [42] Y.-M. Li, Y. Deng, M. Jiang, and D. Wei, "Fast encryption algorithm based on chaotic system and cyclic shift in integer wavelet domain," *Fractal Fractional*, vol. 8, no. 2, p. 75, Jan. 2024.
- [43] A. Bencherqui, M. Amine Tahiri, H. Karmouni, M. Alfidi, S. Motahhir, M. Abouhawwash, S. S. Askar, S. Wen, H. Qjidaa, and M. Sayyouri, "Optimal algorithm for color medical encryption and compression images based on DNA coding and a hyperchaotic system in the moments," *Eng. Sci. Technol., Int. J.*, vol. 50, Feb. 2024, Art. no. 101612.
- [44] P. Kumar, M. Rahman, S. Namasudra, and N. R. Moparthi, "Enhancing security of medical images using deep learning, chaotic map, and hash table," *Mobile Netw. Appl.*, vol. 28, no. 5, pp. 1–15, Sep. 2023.
- [45] K. Panwar, A. Singh, S. Kukreja, K. K. Singh, N. Shakhovska, and A. Boichuk, "Encipher GAN: An end-to-end color image encryption system using a deep generative model," *Systems*, vol. 11, no. 1, p. 36, Jan. 2023.
- [46] N. Mao, X. Tong, M. Zhang, and Z. Wang, "A hyperchaotic image encryption algorithm based on LSTM neural network and lifting wavelet transform," *Phys. Scripta*, vol. 98, no. 7, Jun. 2023, Art. no. 075215.
- [47] Y. Zhao, M. Zheng, Y. Zhang, M. Yuan, and H. Zhao, "Novel dual-image encryption scheme based on memristive cellular neural network and K-means algorithm," *Nonlinear Dyn.*, vol. 112, no. 21, pp. 19515–19539, Jul. 2024.
- [48] M. A. B. Farah, A. Farah, and T. Farah, "An image encryption scheme based on a new hybrid chaotic map and optimized substitution box," *Nonlinear Dyn.*, vol. 99, no. 4, pp. 3041–3064, Mar. 2020.



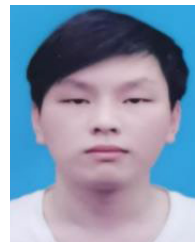
LONGXUAN DAI received the B.S. degree in software engineering from Jiangxi Normal University, Nanchang, China, in 2022, where he is currently pursuing the master's degree in network and information security. His main research interests include image encryption, deep learning, and chaotic systems.



LEI HU (Member, IEEE) received the B.Sc. degree in computer software from Central China Normal University, Wuhan, China, in 2002, the M.E. degree in computer system architecture from Huazhong University of Science and Technology, Wuhan, in 2005, and the Ph.D. degree in computer application technology from Beihang University, Beijing, China, in 2013. She is currently an Associate Professor with the School of Computer and Information Engineering, Jiangxi Normal University, Nanchang, China. Her research interests include image processing, pattern recognition, and machine learning.



LIANGBING CHEN received the B.S. degree in software engineering from Jiangxi Normal University, Nanchang, China, in 2022, where he is currently pursuing the master's degree in network and information security. His current research interests include information security, deep learning, and chaotic systems.



CHENXI WANG received the B.S. degree in software engineering from Nanjing University of Science and Technology Zijin College, in 2022. He is currently pursuing the master's degree in network and information security with Jiangxi Normal University, Nanchang, China. His current research interests include information security, deep learning, log anomaly detection, and chaotic systems.



FENG LIN received the B.S. degree in software engineering from Jiangxi Normal University, Nanchang, China, in 2023, where he is currently pursuing the master's degree in network and information security. His current research interests include information security, deep learning, and chaotic systems.

• • •